



MATHEMATICS
HSSC-I
SECTION - A (Marks 20)
Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Signature should not cross this line

Version No.				
3	2	0	8	1

0	0	●	0	0
1	1	1	1	●
2	●	2	2	2
●	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	●	8
9	9	9	9	9

ROLL NUMBER						

0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
Factorize $z^2 + 169$:	$(z+13)(z-13)$	$(z+13i)$ $(z-13i)$	$(z-13i)$ $(z-13i)$	$(z+13i)$ $(z+13i)$	☐	☐	☐	☐
Argument of $z = 1 + i$ is:	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	☐	☐	☐	☐
Rank of identity matrix is:	Less than its order	Equal to its order	Greater than its order	Undefined	☐	☐	☐	☐
A system is inconsistent if it has:	No solution	Unique solution	Infinite solutions	Repeated solutions	☐	☐	☐	☐
5 th term of A.P for which ($a_1 = 2$ $d = 3$) is:	14	11	13	15	☐	☐	☐	☐
If an A.P. = $a_1, a_2, a_3, \dots$ then H.P. =	$a_1, a_2, a_3, \dots$	$a_1 + a_2 + a_3 + \dots$	$\frac{1}{a_1 + a_2 + a_3 + \dots}$	$\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots$	☐	☐	☐	☒
Coefficient of middle term:	is always 1	is independent of n	Depends on n	is always zero	☐	☐	☐	☐
In mathematical induction first step is:	Hypothesis	Result	Conclusion	Base case	☐	☐	☐	☐
If a cubic polynomial is divided by a linear polynomial (containing same variable), the degree of remainder will be:	0	1	2	3	☐	☐	☐	☐
Using factor theorem If $f(a) = 0$ then:	$f(a)$ is a factor	a is factor	$(x-a)$ is factor	$(x+a)$ is factor	☐	☐	☐	☐
If dot product of two non-zero vectors is zero, then both vectors are:	parallel	equal	mutually perpendicular	in opposite direction	☐	☐	☐	☐
If $\vec{a} = a_1\hat{i} + b_1\hat{j}$ and $\vec{b} = a_2\hat{i} + b_2\hat{j}$ then $\vec{a} \cdot \vec{b}$ is equal to:	$a_1a_2 + b_1b_2$	$a_1a_2 - b_1b_2$	$a_1b_1 + a_2b_2$	$a_2b_1 - a_1b_2$	☐	☐	☐	☐

$x=a$ $(x-a)$



27
7
→



	Question	A	B	C	D	A	B	C	D
13.	Two vectors $\vec{a}$ and $\vec{b}$ will be orthogonal, if the angle θ between them is:	π	$\frac{\pi}{2}$	2π	$\frac{2\pi}{3}$	☐	☐	☐	☐
14.	$\cos(\pi + \beta)$ is equal to:	$\cos \beta$	$-\cos \beta$	$\sin \beta$	$-\sin \beta$	☐	☐	☐	☐
15.	$\cos A \cos B$ is equal to:	$\frac{1}{2}[\cos(A+B) + \cos(A-B)]$	$\cos(A+B)$	$\cos(A-B)$	$\frac{1}{2}[\sin(A+B) + \sin(A-B)]$	☐	☐	☐	☐
16.	Period of $\cos \frac{2x}{5}$ is:	$\frac{5}{2}\pi$	2π	$\frac{2}{5}\pi$	5π	☐	☐	☐	☐
17.	Which of the given functions is even?	$f(x) = x^3$	$f(x) = \sin x$	$f(x) = x$	$f(x) = \cos x$	☐	☐	☐	☐
18.	Maximum value of $y = 3 + 2\sin \theta$ is:	5	4	3	2	☐	☐	☐	☐
19.	Number of ways of arrangement of the letters in the word "GARDEN" are:	480	600	720	840	☐	☐	☐	☐
20.	Number of ways to choose 10 members from 15 are:	360,360	$\frac{15!}{10!}$	$\frac{15!}{5!}$	3003	☐	☐	☐	☐

—1HA-I 26008-32081 (D)—

ROLL NUMBER

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MATHEMATICS HSSC-I

62

Time allowed: 2:35 Hours

Total Marks Sections B and C: 80

SECTION - B (Marks 48)

Q. 2 Solve the following parts.

(12 x 4 = 48)

(Use of the graph paper is not necessary. Candidates can make their own grid on answer book)

(i)	Solve the system: $\begin{aligned} z + 2\omega &= 4i \\ 2z + 5\omega &= 3 \end{aligned}$	04	OR	A force $\vec{F} = 2i + 3j + k$ moves an object through displacement $\vec{d} = i + 2j + 2k$. Find the work done.	04
(ii)	Verify that $\sin 4\theta = 8 \sin \theta \cos^3 \theta - 4 \sin \theta \cos \theta$	04	OR	Evaluate $\frac{(n-1)!}{(n-3)!}$	04
(iii)	Find the volume of a tetrahedron formed by vectors: $\vec{a} = 2i + j, \vec{b} = i + 3j + k, \vec{c} = j + k$	04	OR	Find the maximum and minimum values of: $y = \frac{1}{2 + 3 \sin(2x + 1)}$	04
(iv)	Find the value(s) of K if $x - 1$ is a factor of: $f(x) = x^3 + K^2 - 2K + 1$	04	OR	Find the sum of 6 terms of sequence: $1 + \frac{1}{2} + \frac{1}{4} + \dots$	04
(v)	Without drawing graph determine the amplitude, period and range of: $y = 2 \sin(3x + 1)$	04	OR	Find the angle between vectors: $\vec{a} = 4i + j - 8k, \vec{b} = i + 4j + k$	04
(vi)	Find the 10 th term of an A.P with $a = 3$ and the 7 th term of it is 15.	04	OR	Using fundamental law of trigonometry find the value of $\cos 15^\circ$. (do not use calculator)	04
(vii)	Find the rank of matrix: $\begin{bmatrix} 3 & -4 & 2 \\ 2 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$	04	OR	Express $2 \sin 12\theta \cos 6\theta$ as a sum or difference.	04
(viii)	The 3 rd term of a geometric sequence is 16 and the 5 th term is 64. Find the common ratio and 1 st term. Also write the sequence for positive values.	04	OR	Find the inverse of given matrix if possible: $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 2 \\ 2 & 1 & 4 \end{bmatrix}$	04
(ix)	Determine the last two digits of $(13)^{10}$.	04	OR	Sketch the graph of $y = 3 \sin x$ for $-\pi \leq x \leq \pi$	04
(x)	Convert $z = 1 + i$ into polar form and then find z^2 using polar representation.	04	OR	Find r if ${}^{54}P_r : {}^{56}P_{r+3} = 1 : 30800$	04
(xi)	Use Binomial Theorem to find the remainder when 8^{205} is divided by 48.	04	OR	Factorize the polynomial: $x^3 - 4x^2 + x + 6$	04
(xii)	For what value of λ the given matrix is singular? $\begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & \lambda \\ 3 & 1 & 2 \end{bmatrix}$	04	OR	Use mathematical induction to prove: $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$	04

SECTION - C (Marks 32)

Note: Solve the following Questions.

(4 x 8 = 32)

(Use of the graph paper is not necessary. Candidates can make their own grid on answer book)

Q.3	a. A company's profit over the last few years can be modeled by the polynomial $P(x) = x^3 + 6x^2 - x - 6$. If x represents years since 2010, calculate the company's profit in 2012 by dividing $P(x)$ and $x - 2$ finding the remainder. b. Verify your answer by long division.	08	OR	If $\vec{a} = 3\hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$, then a. Find $\vec{a} \times \vec{b}$ b. Find $(\vec{a} \times \vec{b}) \cdot \vec{a}$ and verify that $\vec{a} \times \vec{b} \perp \vec{a}$ c. Find $(\vec{a} \times \vec{b}) \cdot \vec{b}$ and verify that $\vec{a} \times \vec{b} \perp \vec{b}$	08
Q.4	Find the term independent of x in the expansion of $\left(2x - \frac{1}{x}\right)^{12}$. Also find the 10 th term.	4+4	OR	Prove fundamental law of trigonometry: $\cos(A - B) = \cos A \cos B + \sin A \sin B$ Diagram is necessary.	08
Q.5	Solve the system: $\begin{aligned} x + y + z &= 6 \\ 2x + 3y + z &= 10 \\ x + 2y + 3z &= 11 \end{aligned}$	08	OR	The height of seat on a Ferris wheel is modeled by the equation $h(t) = 10 + 8 \sin \frac{\pi}{6} t$ a. Find the period of motion b. Determine the maximum and minimum height c. At what time is the height maximum?	3+2+3
Q.6	a. For the sequence 2, 5, 8, 11, ... find the general term. Also find the sum of first 20 terms. b. Find the sum of the n terms of the series $1.2 + 2.3 + 3.4 + \dots$	4+4	OR	A box contains 7 red, 5 blue and 4 green balls. In how many ways can 5 balls be selected if at least 2 are red?	08